CNN-LSTM Remaining Duration Prediction and Workflow-Informed Tool Detection on Chloec80

By: Nate Chu

Abstract

Studying surgical workflow on the Cholec80 laparoscopic dataset. In Task A, utilizing a CNN–LSTM regressor to predict phase timing from 1 fps video, facilitating a comparison of temporal windows (K=60 vs K=120). The best K=60 model achieves 215.97 s MAE for current-phase end-time prediction with 29.2% within ±60 s, while K=120 does not improve performance (263.49 s MAE, 27.7% within ±60 s). Building on these temporal features, Task B evaluates whether predicted workflow timing can enhance tool presence by concatenating Task A timing features with visual features. The time-aware model improves test performance against a visual-only baseline (mAP 0.9102 → 0.9182, Micro-F1 0.8852 → 0.8918), suggesting workflow timing provides complementary context for tool recognition.

1. Introduction

Laparoscopic surgery is a form of minimally invasive surgery performed in the abdominal section. Specifically, cholecystectomy is an operation to remove the gallbladder, which may need to be extracted if gallstones begin to form or cause stomach pains. The majority of cholecystectomy is performed by laparoscopy [5], presenting a vast source of video data, allowing machine learning algorithms to learn and be leveraged for surgical workflow.

Across multiple studies, its application for phase recognition in surgical procedures seems the most impactful [1, 3, 6]. Enabling context-aware systems for intraoperative assistance, surgical training and operating room optimization. Accurate modelling of surgical workflow is particularly relevant in laparoscopic cholecystectomy, where complications such as bile duct injury remain a significant clinical risk[4]. Timely and reliable contextual awareness may therefore act as a “second pair of eyes”, supporting surgeons by improving situational awareness and patient safety [4].

Our primary goal was to determine if adding temporal context to a CNN-LSTM model improves surgical workflow understanding. We first investigated the optimal context window for time prediction (Task A), specifically whether a longer history improves accuracy. We then used these timing predictions as an auxiliary "prior" to see if a tool detector could be made more robust by "knowing" the current stage of the surgery (Task B), comparing this against a standard visual baseline.

2. Methods

2.1 System Overview (CNN + LSTM)

A CNN–LSTM architecture is adopted to model the spatial-temporal structure of laparoscopic surgical workflow. Laparoscopic video exhibits high visual variability at the frame level due to camera motion, occlusion, smoke, and tissue deformation, while the underlying workflow evolves according to a temporally ordered sequence of phases with strong duration constraints. Separating spatial and temporal modelling therefore provides an effective inductive bias for this domain [1, 3].

The convolutional neural network (CNN) serves as a learned visual feature extractor, mapping raw image frames to compact representations encoding instrument appearance, anatomical context, and global scene structure. Compared to handcrafted visual features, CNN representations are better suited to capture subtle cues [1] such as partial tool visibility and changes in surgical field configuration, which are difficult to model explicitly, yet highly relevant to workflow progression. ImageNet pretraining further improves generalisation by providing robust low-level feature representations under limited medical data availability [1, 6].

Temporal dependencies are modelled using a Long Short-Term Memory (LSTM) network, as surgical procedures evolve sequentially with strong ordering constraints and variable phase durations. LSTMs have been widely used in surgical phase recognition and workflow modelling [1] due to their ability to integrate information over extended temporal horizons.

A unidirectional LSTM is used instead of a bidirectional variant to preserve causal, online inference. While bidirectional LSTMs can improve offline phase recognition by exploiting future frames, they are unsuitable for real-time surgical assistance, where predictions must rely only on past and present information. Prior surgical workflow models targeting deployable systems therefore adopt unidirectional temporal models [2, 3, 6], including CNN–LSTM architectures for phase recognition and remaining surgery duration prediction.

Using a unidirectional LSTM ensures that predicted timing features used in Task B are generated causally and can be consumed frame-by-frame without access to future context, maintaining methodological consistency and clinical plausibility

2.2 Task A: Temporal Workflow

Task A focuses on predicting clinically meaningful temporal quantities from laparoscopic video, including the remaining time of the current phase. Including the remaining time of the current phase and the expected start and end times of all future phases. These estimates provide explicit workflow context and form the basis for the time-aware models in task B.

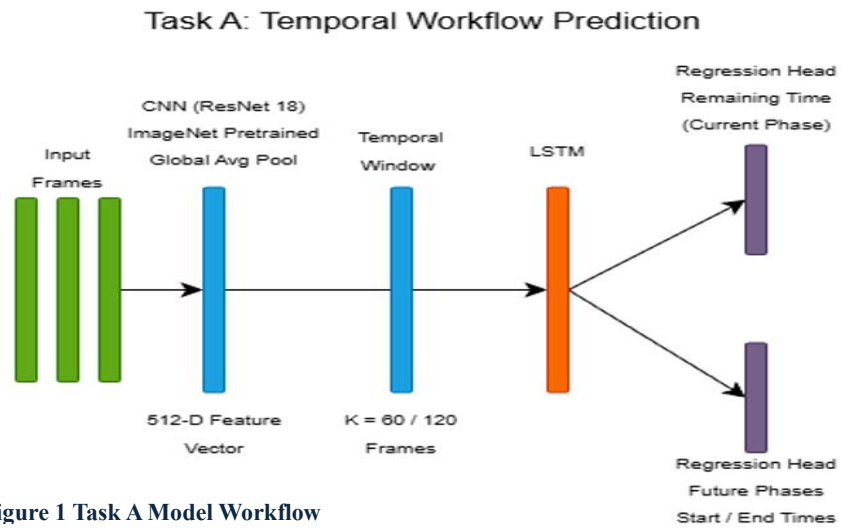


Figure 1 Task A Model Workflow

2.2.1 Spatial Feature Extraction

This architecture is adopted to jointly model spatial appearance and temporal workflow structure. The CNN learns discriminative visual representations of instruments, tissue, and camera motion, while the LSTM captures sequential dependencies arising from the ordered and duration-constrained nature of surgical phases. Spatial feature extraction utilizes an ImageNet-pretrained ResNet-18 backbone (as shown in Figure 1). This initialization improves convergence and generalization on the relatively limited laparoscopic dataset.

2.2.2 Time Regression Target

The LSTM hidden state is mapped to temporal predictions using a lightweight two-layer MLP. The regression head outputs a 14-dimensional vector, corresponding to the remaining time until the start and end of each of the seven surgical phases.

Target times are expressed in seconds and log-transformed as $y = \log(1 + t)$ to compress the wide range of surgical durations. This transformation emphasises accuracy in short-term, clinically relevant predictions, such as the end of the current phase.

2.2.3 Training Strategy and Loss

A 60/10/10 split is adopted for the dataset; these splits balance the need for sufficient training data to learn temporal workflow patterns with reliable model selection and unbiased evaluation. Validation videos are used for checkpoint selection and hyperparameter tuning, while the test set is held out for final performance reporting.

Training is performed in two stages. First, the CNN backbone is frozen (until epoch 3) while the LSTM and regression head are trained. The full network is then fine-tuned end-to-end using a reduced learning rate, allowing gradual adaptation of pretrained visual features to the laparoscopic domain.

A masked, weighted Smooth L1 loss is used for regression. Loss is computed only for phases present in the current procedure, including the end time of the current phase and the start and end times of future phases. The end of the current phase is weighted more heavily (4.0) than future phase estimates (1.0) to reflect its greater clinical relevance.

Optimisation is performed using AdamW with learning rates of 1×10^{-4} and 1×10^{-5} for the two training stages respectively. A batch size of 2 is used due to GPU memory constraints. Each training sample consists of a fixed temporal window of 60 frames (60 s at 1 fps), providing sufficient context while remaining computationally feasible.

2.3 Task B: Time-Aware Surgical Vision Model

Task B tests whether predicted workflow timing improves tool presence detection. We compare a visual-only baseline to a time-aware model that concatenates Task A timing features with the visual representation; all other architecture and training settings are kept identical.

2.3.1 Pre-computation of Temporal Features

Temporal context is obtained from the trained Task A model, which predicts clinically meaningful temporal quantities from laparoscopic video. Using the selected Task A checkpoint (chosen based on validation performance), temporal predictions are

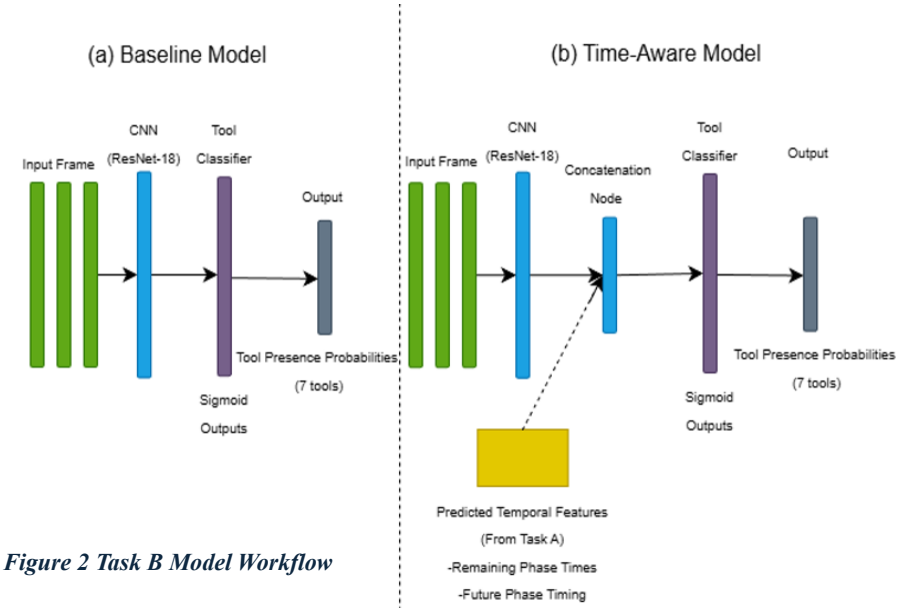


Figure 2 Task B Model Workflow

generated offline for all frames in the training, validation and test splits used in Task B.

For each video frame, Task A model outputs predicted temporal features including: 1. Estimated remaining time to end of current surgical phase and 2. Estimated start and end times of upcoming surgical phases. These predictions are converted back to seconds and stored as per-frame feature vectors.

2.3.2 Baseline Vision Model

The baseline model operates solely on visual information extracted from laparoscopic video frames. A CNN-based visual encoder identical to that used in Task A is employed to extract spatial feature representations from individual frames. These features are passed to a task-specific prediction head, to produce per-frame predictions.

2.3.3 Time-Augmented Model

The time-aware model concatenates the per-frame timing feature vector from Task A with the CNN visual feature vector before the tool-classification head.

3. Experiment

3.1 Dataset and Preprocessing

All experiments are conducted on the Cholec80 dataset, which consists of 80 laparoscopic cholecystectomy procedures annotated with surgical phase and tool presence labels. The original videos are recorded at 25 frames per second and are uniformly downsampled to 1 frame per second to reduce temporal redundancy and computational cost.

Phase annotations are provided at the original 25 fps resolution, resulting in multiple identical phase labels within each one-second interval. Since phase labels remain constant over this interval, the annotations are collapsed by assigning a single-phase label to each sampled image frame. Tool annotations are defined on the same 25 fps timeline but are provided at 25-frame intervals, with each entry representing tool presence over one second.

3.2 Metrics

3.2.1 Metrics (Task A)

Task A is evaluated using regression-based metrics that quantify the accuracy of predicted temporal quantities. Predictions are converted back to seconds by inverting the logarithmic transformation applied during training.

The primary metric is the mean absolute error (MAE) for the predicted end time of the current surgical phase:

$$MAE = \frac{1}{N}\sum_{i=1}^{N} |\widehat{t_i} - t_i| \quad (1)$$

Where $\widehat{t_i}$ and t_i represent the predicted and ground-truth times, respectively.

In addition, a **tolerance-based accuracy** metric is reported, defined as the percentage of predictions falling within ±60 seconds of the ground truth.

$$|\widehat{t_i} - t_i| \leq 60 \text{ seconds} \quad (2)$$

This metric reflects whether predicted timings are sufficiently precise to support intraoperative decision-making as used in prior surgical workflow evaluation [1, 3].

3.2.2 Metrics (Task B)

Task B is evaluated as a multi-label tool presence detection problem. Model output is activated probabilities for each of the seven surgical tools and are compared against the binary ground truth labels.

Performance is quantified by using micro-averaged F1-score at a probability threshold of 0.5:

$$F1_{micro} = \frac{2TP}{2TP + FP + FN} \quad (3)$$

Where, TP (Total True Positives) and FP (Total False Positives) and FN (Total False Negatives). Micro-F1 is well suited to this task as it accounts for class imbalance rather than per-class performance. As there are tools which appear far less frequently than others.

Additionally, tool detection is assessed using Average Precision (AP) for each tool. AP is computed as the area under the precision-recall curve obtained by varying the detection threshold across the full range of predicted probabilities.

$$AP = \int_0^1 P\,(r)\,dr \quad (4)$$

Where $P\,(r)$ denotes precision as a function of recall r.

Mean Average Precision (mAP) is reported as the mean AP across all $K = 7$ tools:

$$mAP = \frac{1}{K}\sum_{k=1}^{K} AP_k \quad (5)$$

AP and mAP measure the model's ability to rank frames with true tool presence ahead of negatives, independent of a fixed operating threshold. This is particularly relevant in surgical applications, where the optimal sensitivity-specificity trade-off may vary across clinical contexts.

3.3 Evaluation and Comparison

3.3.1 Evaluation and Comparison (Task A)

For Task A, evaluation focuses on the accuracy of temporal workflow predictions from laparoscopic video. The model is evaluated on the held-out test set using the selected checkpoint based on validation performance. During inference, each procedure is processed sequentially using a sliding temporal window, producing predictions of the remaining time to the end of the current phase as well as the estimated start and end times of all future phases.

Performance is assessed using methods described in section 3.2.1. Errors are computed in seconds after inverting logarithmic transformations applied during training. Metrics are reported separately for the end time of the current phase and for the start and end times of all future phases within the procedure, allowing assessment of both short-term and long-term temporal reasoning.

3.3.2 Evaluation and Comparison (Task B)

Task B is evaluated by comparing two models: a baseline model relying solely on visual features and a time-aware variant that incorporates predicted temporal information from Task A. Evaluation is performed on the test set using Micro-F1 and mAP (Section 3.2.2).

For the time-aware model, temporal features corresponding to each frame are obtained from Task A predictions and concatenated with visual feature representations prior to the prediction head. All other architectural and optimization settings are kept identical to the baseline model.

Performance differences between the two models therefore directly reflects the contribution of explicit temporal information. This comparison assesses whether predicted workflow timing improves tool presence detection reliability beyond visual cues.

A non-parametric paired bootstrap test was selected because tool detection metrics are computed over correlated frames within each video and do not satisfy independence assumptions required by parametric tests. Resampling at the video level preserves temporal structure while providing a distribution-free estimate of performance differences.

4. Results

4.1 Task A: Effect of Temporal Context on Time Prediction

Shown in Figure 3 it compares validation MAE for temporal context lengths of K = 60 and K = 120 across 6 epochs. Both configurations exhibit rapid convergence, with no sustained performance advantage observed for the longer temporal window. Based on validation performance, the K = 60 model at epoch 6 and for K = 120 at epoch 1 was selected for final evaluation.

Test-set results are summarized in Table 1. The two temporal configurations achieved comparable MAE and tolerance-based accuracy, with no consistent improvement observed when increasing the temporal context from 60 to 120 frames. This indicates diminishing returns from longer temporal histories consistent with prior workflow models[6].

These results suggest that shorter temporal windows are sufficient to capture the relevant workflow dynamics, supporting lower latency and computational cost for real-time deployment without compromising predictive accuracy.

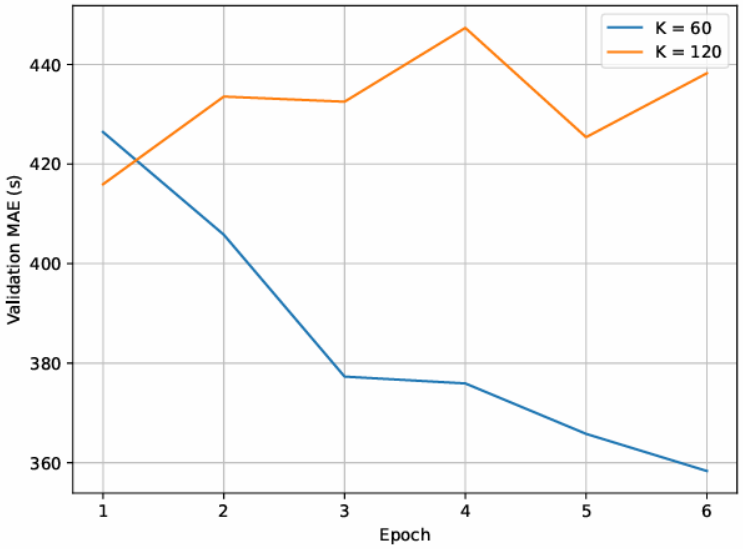


Figure 3 Validation MAE vs Epoch for K=60 & K=120

Table 1. Test-set performance for Task A across temporal windows

Temporal Window (K)	MAE (s) ↓	Predictions Within ±60 s (%) ↑
60 (Epoch 6)	215.97	29.2
120 (Epoch 1)	263.49	27.7

4.2 Task B: Impact of Predicted Time On Tool Detection

Overall tool detection performance for baseline and time-aware models is reported in Table 2, while Figure 4 visualises the corresponding difference in mean Average Precision (mAP). Incorporating predicted surgical timing as an auxiliary input leads to a modest but consistent improvement in mAP on both validation and test sets compared to the baseline model.

Although the absolute performance gain is small, the improvement is consistent across evaluation splits and is also reflected in Micro-F1 scores, suggesting that temporal information provides complementary contextual cues beyond visual appearance alone. These findings support the hypothesis that predicted surgical timing can serve as useful context prior for tool presence detection.

Such gains are relevant in workflow-aware systems, particularly for tools whose usage is strongly phase-dependent.

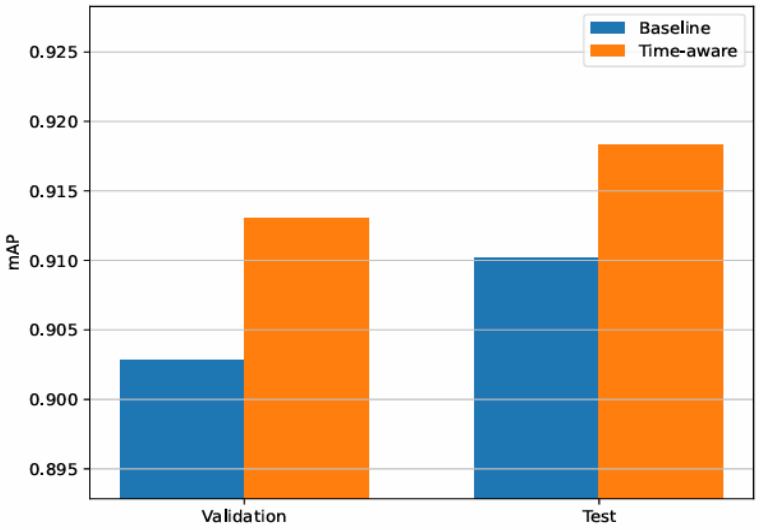


Figure 4 Difference in mAP between Baseline and Time-aware model

Table 2. Tool detection performance on test set

Model	Split	mAP ↑	Micro-F1↑
Baseline	Test	0.9102	0.8852
Time-Aware	Test	0.9182	0.8918

4.3 Qualitative and Quantitative analysis

Formal statistical significance testing was not performed for Task A, as the evaluation concerns absolute regression accuracy of a single model rather than a paired comparison between competing methods. Performance is therefore reported directly using MAE and tolerance-based accuracy, which are standard for temporal workflow prediction tasks.

It should be noted that due to computational constraints, each model configuration was evaluated based on a single training run. Consequently, the reported metrics represent specific performance snapshots rather than averaged results across multiple initializations.

4.3.1 Task A Analysis

The discrepancy between the absolute MAE (~216 s) and the low tolerance-based accuracy (29.2% within ±60 s) suggests the model functions better as a general workflow progress signal than a precise countdown clock. From a clinical perspective, this indicates the system is reliable for high-level OR coordination but may not yet support tasks requiring second-to-second precision.

The finding that the K=120 model did not outperform the K=60 configuration suggests that "more data" isn't always better here. It seems that adding older frames effectively introduces 'noise': what happened two minutes ago is often irrelevant to the current phase, especially near a transition. For real-time use, this is a positive finding, as shorter windows are less computationally expensive and provide better predictive focus.

4.3.2 Task B Analysis

Time-aware model demonstrates higher AP for several phase-specific instruments, most notably the bipolar and scissors, while tools that appear across multiple phases, such as the grasper and hook exhibit smaller changes. A small decrease in AP is observed for some tools, including the clipper, indicating the benefits of temporal context are not uniform across all instrument types. This pattern suggests that predicted timing function as a weak prior: it helps most when tool usage is strongly phase coupled [1,6] and visually ambiguous.

Table 3. Tool specific AP test metric

Tool	Baseline AP	Time-aware AP
Grasper	0.8635	0.8499
Bipolar	0.8487	0.9065
Hook	0.9930	0.9907
Scissors	0.8550	0.8702
Clipper	0.9941	0.9655
Irrigator	0.8554	0.8766
Specimen bag	0.9613	0.9682

Qualitatively, this supports a workflow-aware interpretation: timing features likely improve detection by discouraging tool activations at implausible points in the procedure, while tools that appear broadly across phases (or have already saturated performance) gain less and may fluctuate slightly.

4.4.1 Two-sided bootstrap test (Task B)

To assess whether the observed performance improvements from incorporating predicted workflow timing are robust, paired statistical significance testing is performed on the test set.

A paired video-level bootstrap test is used to compare the time-aware and baseline models. Videos are sampled with replacement over 2000 bootstrap replicates, and for each replicate the paired difference in mAP and Micro-F1 between models is computed. This approach accounts for temporal correlation between frames within the same video and avoids inflating the effective sample size.

Table 4. Paired Video-Level Bootstrap Comparison (n = 10 test videos).

Metric	Mean Δ (Timed − Baseline)	95% CI	p-value
mAP	+0.0092	[−0.0013, +0.0179]	0.11
Micro-F1	-0.0072	[−0.0116, −0.0034]	<0.0001

Table 4 reports the results of a paired video-level bootstrap test comparing the baseline and time-aware models on the test set. Incorporating predicted workflow timing resulted in a positive but non-significant improvement in mAP, while Micro-F1 decreased significantly at a fixed decision threshold. These results indicate that timing information affects ranking-based and threshold-based performance differently.

5. Discussion

For phase timing, the closest comparable idea is Twinanda et al offline phase-boundary localisation tolerance [3], where they report 89% of phase boundaries detected within 30 s. Our tolerance-based accuracy for “end of current phase” is lower (29.2% within ±60 s, MAE ≈ 216 s for K=60), highlighting that continuous remaining-time estimation is substantially harder and more sensitive to inter-surgeon variability in phase duration.

Phase duration can differ substantially between surgeons and cases (e.g., slow, meticulous dissection versus rapid straightforward cases). A model trained on absolute time may therefore miscalibrate: for a slow surgeon it may systematically under-estimate remaining time, while for a fast surgeon it may over-estimate. Two practical mitigations are (i) pace normalisation, predicting relative progress within phase (or conditioning timing on elapsed phase time) rather than absolute seconds, and (ii) uncertainty-aware timing, outputting a distribution or confidence interval so downstream tool detection can use timing as a soft prior rather than a hard constraint. More robust generalisation could also be achieved by stratifying evaluation by procedure duration (fast/median/slow terciles) and reporting whether timing features help consistently across strata.

Previous EndoNet paper published by Twinanda et al. evaluates tool presence on Cholec80 using AP/mAP and reports mean AP ≈ 81% [1]. Using the same AP/mAP family of metrics, our models achieve mAP = 0.9102 (baseline) and 0.9182 (time-aware) on the test split. Numerically, these values exceed Twinanda et al. reported mean AP; however, this should be interpreted cautiously rather than as a definitive “better model”, because Twinanda et al. result is tied to its specific dataset partition, training setup (multi-task network with different downstream smoothing for phases), and evaluation subset, which are not matched here. A safer conclusion is that a relatively lightweight CNN–temporal pipeline can reach high tool-presence AP on this evaluation, and adding explicit timing provides a small additional gain without changing the visual backbone.

As shown in Table 4 Interestingly, the time-aware model improved the mAP but slightly hurt the Micro-F1. This indicates that while the timing features help the model "rank" frames better (putting the right tool at the top), timing might be throwing off the model's confidence scores. By injecting timing data, it may have shifted the probabilities in a way that makes a fixed 0.5 threshold less effective. Given that there were only 10 test videos, it’s clear that the timing signal acts as a "soft" hint rather than a hard rule for detection.

6. Conclusion

This project shows that while timing prediction in surgery is difficult due to surgeon variability, it still provides a valuable context signal for tool detection. A 60-second history was the "sweet spot" for our CNN-LSTM, as longer histories seemed to dilute the relevant signals. While the gains in tool detection were modest, the results suggest that "temporal priors" are a viable way to improve surgical AI. Future work should focus on predicting relative progress (e.g., % of phase complete) rather than absolute seconds to better handle different surgical paces.

Bibliography

[1] Twinanda, A.P., Shehata, S., Mutter, D., Marescaux, J., de Mathelin, M., Padoy, N.:
EndoNet: A deep architecture for recognition tasks on laparoscopic videos. *IEEE Transactions on Medical Imaging* **36**(1), 86–97 (2017). https://doi.org/10.1109/TMI.2016.2593957

[2] Twinanda, A.P., Yengera, G., Mutter, D., Marescaux, J., Padoy, N.:
RSDNet: Learning to predict remaining surgery duration from laparoscopic videos without manual annotations. *IEEE Transactions on Medical Imaging* **37**(5), 1069–1078 (2018).

[3] Yu, T., Mutter, D., Marescaux, J., Padoy, N.:
Learning from a tiny dataset of manual annotations: A teacher–student approach for surgical phase recognition. In: *Proc. IPCAI*, LNCS, Springer (2020).
(arXiv:1812.00033)

[4] Lalys, F., Jannin, P.:
Computer assisted minimally invasive surgery: Is medical computer vision the answer to improving laparosurgery? *Computerized Medical Imaging and Graphics* **36**(6), 425–434 (2012).
https://doi.org/10.1016/j.compmedimag.2012.07.003

[5] National Health Service (NHS):
Gallbladder removal (cholecystectomy).
https://www.nhs.uk/tests-and-treatments/gallbladder-removal/what-is-gallbladder-removal/
Accessed: Jan 2026

[6] Jin, Y., Dou, Q., Chen, H., Yu, L., Qin, J., Fu, C.W., et al.:
SV-RCNet: Workflow recognition from surgical videos using recurrent convolutional network.
IEEE Transactions on Medical Imaging 37(5), 1114–1126 (2018).